

Lecture 8

Sept 17

inverse prob...

Learning objectives

- Distinguish calculating the probability of observations given a true value from using observations to assess possible true values, and explain why the latter requires an additional inferential step.
- Calculate and interpret conditional probabilities from a two-way table, explain how conditioning changes the relevant population.
- Use conditional probability and the law of total probability to explain Laplace's inverse-probability rule for equally likely causes, and identify the conditional independence assumption in his argument about repeated observations.
- Describe Laplace's criteria for an error curve and evaluate the assumptions behind his choices.

What we have ...

$X \sim \underline{P_0} \leftarrow \text{distribution}$

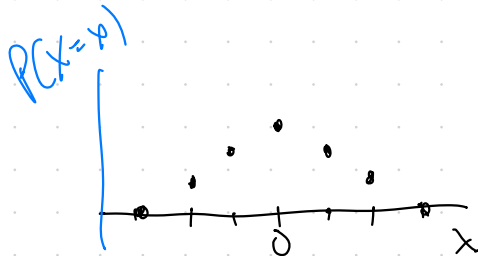
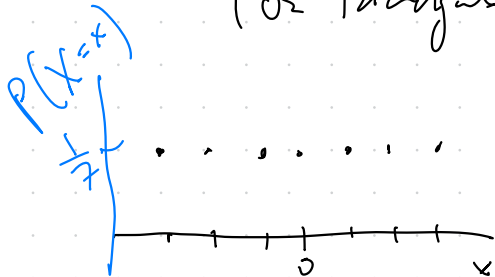
$\underline{E[X]} = \sum x p(X=x)$ expectation

$\underline{\text{Var}[X]} = E[(X - E(X))^2]$ variance

$\underline{\text{SD}[X]} = \sqrt{\text{Var}[X]}$ standard deviation

Simpson + Bayes draw us to studying the error.

Simpson's errors are symmetric around zero, and uniform (or triangular) and have a "maximum"



Daniel Bernoulli

1700-1782, Born in Groningen, died in Basel

Son of Johann, nephew of Jacob (the bickering brothers)

He just wanted to please his father...

- ① places 1st in a competition in the Paris Academy of Science.



- ② Writes a book called Hydrodynamica, signs it D. son of J. in 1738.



Daniel's Contributions (to us)

- Widely regarded as deriving the idea of maximum likelihood — which isn't formalized until 100 years later!
- Essentially, if you have data $x_1, \dots, x_n$ and a model for the data $p(x)$ with a parameter θ isn't there a parameter value that makes the data most likely? $f(\theta) = p(x | \theta) \quad \max_{\theta} f(\theta) \quad \text{vs} \quad p(\theta | x)$
- DRAMA here — Euler published a critique saying outliers would be a problem.

Laplace and cause and effect.

What is a cause?

- an action or event that leads to something else
- an event that generates a probability

For Laplace + co "cause" leads to something else but is not action

Jacob Bernoulli and De Moivre can't figure out the "inverse"

$P(|\frac{X}{N} - \frac{a}{a+b}| > \varepsilon)$ is bounded by a function of a, b
but a, b are unknown.

Laplace translates the problem to errors

random error ε

Observation O

True location P

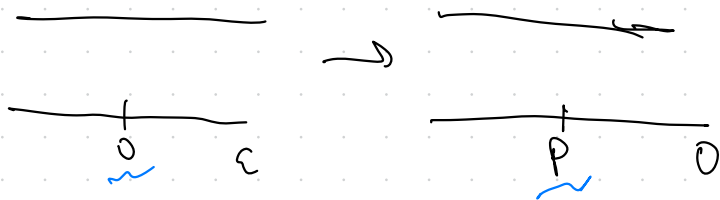
$$O = P + \varepsilon$$

effect cause

P is fixed then distribution of O

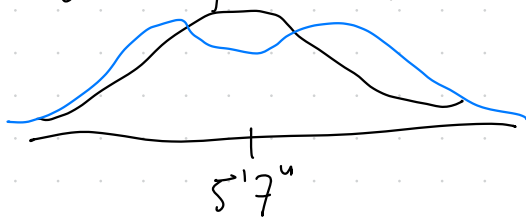
$$P = O - \varepsilon$$

let O be fixed then I have
distribution for P



What is a ~~conditional~~ probability?

Heights of 18 year olds



$$P(H < 5'7")$$

Conditional prob.

Heights of 18 year old who are women
conditioning set

Restrict the population
about which we are
talking about

$$P(H < 5'7" \mid \text{women})$$

↑
"conditional"

Who survives a sea passage?

	Survived	Did not survive	Total
Passenger	36	24	60
Crew	12	28	40
Total	48	52	100

① Pick a person at random. What is $P(\text{survived})$?

$48/100$

② The person is a passenger. Is the prob higher, lower, the same?

higher

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- ① Pick a person at random. What is $P(\text{survived})$?
 $\frac{48}{100}$
- ② The person is a passenger. Is the prob higher, lower, the same?
- ③ What is $P(\text{survived} \mid \text{it is a passenger})$?
 $\frac{36}{60}$
- ④ What is $P(\text{passenger} \mid \text{they survived})$?
 $\frac{36}{48}$

Formal definition

Suppose there are N equally likely cases

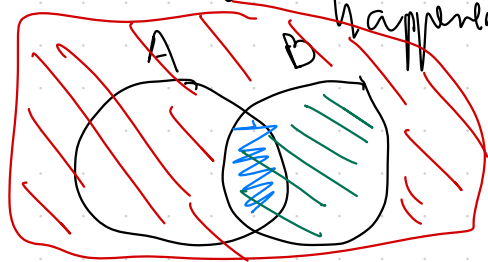
The event B contains N_B cases $P(B) = N_B / N$

The event " A and B " ($A \cap B$) contains $N_{A \cap B}$ cases

$$P(A \cap B) = N_{A \cap B} / N$$

If B is known to have happened then there are $N - N_B$ cases that are not possible

Among the possible cases what are the chances that A



happened?

$$P(A|B) = N_{A \cap B} / N_B$$

$$= \frac{N_{A \cap B} / N}{N_B / N} = \frac{P(A \cap B)}{P(B)}$$

Laplace's method through conditional probabilities.

If an event can be produced by a number n of different causes, then the probabilities of these causes given the event are to each other as the probabilities of the event given the causes, and the probability of the existence of each of these is equal to the probability of the event given that cause, divided by the sum of all the probabilities of the event given each of these causes. (Laplace, 1774, p. 623)

$$P(E|A_i) \xrightarrow{?} P(A_i|E)$$

$$P(A_i|E) = \frac{P(E|A_i)P(A_i)}{\sum_j P(E|A_j)P(A_j)}$$

Manual derivation

The event E happens two times.

E_1 is the first occurrence. E_2 is the second

$A_1, \dots, A_n$ are "causes" and $\sum_i P(A_i) = 1$ and they are independent

Big (important) assumption: E_1 is independent of E_2 given A_j .

Need $P(E_2 | E_1) = \sum_i P(E_2 \cap A_i | E_1)$ law of total prob.

$$= \sum_i P(E_2 | A_i \cap E_1) P(A_i | E_1) \quad \text{def'n of conditional prob.}$$

$$= \sum_i P(E_2 | A_i) P(A_i | E_1) \quad \text{by conditional indep.}$$

$$P(E_2|E_1) = P(E_1 \cap E_2) / P(E_1) \quad \text{by definition of conditional prob.}$$

$$P(E_1 \cap E_2) = \sum P(E_1 \cap E_2 | A_i) P(A_i) \quad \text{by definition of conditional prob.}$$

$$= \sum P(E_1 | A_i) P(E_2 | A_i) P(A_i) \quad \text{by independence}$$

$$= \sum P(E | A_i) P(E | A_i) P(A_i) \quad \text{because } E_1, E_2 \text{ are just independent copies of the generic event } E$$

$$= \sum P(E | A_i)^2 P(A_i)$$

$$= \frac{1}{n} \sum P(E | A_i)^2 \quad \text{by } P(A_i) = \frac{1}{n} \text{ assumption}$$

$$P(E_1)$$

$$= \sum P(E_1 \cap A_i) \quad \text{by total probability}$$

$$= \sum P(E_1 | A_i) P(A_i) \quad \text{by definition of conditional prob.}$$

$$= \frac{1}{n} \sum P(E | A_i) \quad \text{by } P(A_i) = \frac{1}{n} \text{ assumption}$$

$$\sum P(E | A_i) P(A_i | E) = \frac{\cancel{\frac{1}{n} \sum P(E | A_i)^2}}{\cancel{\frac{1}{n} \sum P(E | A_i)}}$$

This process can be repeated with E occurring k times to get $n-1$ equations $\rightarrow$

Solving yields $P(A_i|E) = \frac{P(E|A_i)}{\sum_j P(E|A_j)}$

Modern framing

$$P(A_i|E) = \frac{P(E|A_i) P(A_i)}{\sum_j P(E|A_j) P(A_j)}$$

note the "missing" $P(A_i)$
because of equiprobability
of causes assumption

Laplace's tasks

- ① What curve should be used to describe the probability?
- ② Given the curve, what should be the mean?

The principle lets him do

$$P(\text{causes} \mid \text{effects}) \propto P(\text{effects} \mid \text{causes})$$

$$P(\text{parameter} \mid \text{data}) \propto p(\text{data} \mid \text{parameters})$$

What's $a \propto b$?

$$a = b \cdot c$$

where c doesn't depend on b .

Choose your curve!

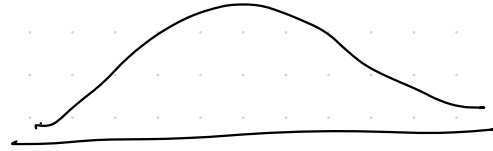
① Symmetric about zero

② Decrease as we move away from zero

③ Error should have a prob. distn

Choose your "mean"!

$P(\text{Parameters} | \text{data})$



- the fun
myer
- ① Equally probable that the true value is
before or after my chosen mean
mean of probability
posterior median
 - ② minimize the sum of errors
astronomical mean
minimizes posterior expected loss
under absolute loss

One curve to rule them all?

① Symmetrical

② Decreases

Rough sketch of his approach

He wanted $\phi(x)$ and $d\phi(x)/dx$ to decrease!